The Readiness Option

Conservation Priority over Space and Time*

Diana Weinhold† Lykke E. Andersen‡

November 2026

Abstract

We show that when conservation builds *readiness* – administrative experience, public legitimacy, and political capacity that make later protection elsewhere feasible or less costly – conserving ecologically valuable land can be optimal well before it faces immediate threat. In a real-options model, some parcels are worth securing but the timing of threats is uncertain; a favorable acquisition window may close before conversion is imminent. We show that greater uncertainty about the stakes of later conflicts can raise or lower the marginal value of readiness, depending on the agency's initial capacity. For agencies with sufficiently low initial capacity, greater uncertainty can bring protection forward when the acquisition window is sufficiently fragile. Readiness can also deter challenges to protected land before they materialize, compounding the difficulty of measuring avoided loss for protection that precedes threat. Finally, for comparable development projects, siting priority rules that target the most ecologically valuable land ensure that development deterred or defeated at the protected site does no more damage elsewhere; local readiness-enhancing protection can thus improve conservation of the wider landscape.

Keywords: conservation priority, conservation timing, dynamic reserve selection, real options, institutional capacity, additionality, protected areas.
JEL: Q57, Q58, D81, C61, H43, Q24.

*We thank Eduardo Forno, Ben Groom, Charlie Palmer, Ben Balmford, and Sabrina Eisenbarth for the lively discussions that motivated this paper, and the Natural Environment Research Council (NERC) for funding. An AI-use disclosure statement appears in the Appendix.

†Department of International Development, London School of Economics. Email: d.weinhold@lse.ac.uk.
‡Universidad Privada Boliviana - SDSN Bolivia. Email: lykkeandersen@upb.edu

1 Introduction

When should a conservation planning agency (henceforth the agency) protect valuable land that is not yet under immediate threat? A parcel may face little conversion pressure today, but its future uses and the opportunities to secure it are uncertain. Waiting can be sensible: it preserves scarce resources and allows better information to arrive. But waiting can also be costly – land prices may rise, political opposition may strengthen, or the opportunity to acquire the parcel may otherwise disappear. We add an institutional reason for pre-emption: protection in one location can also increase the agency's experience, relationships, and political constituency that support its capacity to maintain the protection and to conserve elsewhere. We call this capacity *readiness*. Its benefits arise at other sites, at later dates, and as challenges that never occur, so they are easy to miss in a short-horizon, site-level evaluation of impact. We show how incorporating readiness into an optimal timing decision changes *when* land should be protected, how the effect of uncertainty on that timing depends on the agency's initial capacity, and how siting priority – *where* the agency protects – shapes what protection achieves across the wider landscape.

A case in point is the Tucabaca reserve in eastern Bolivia, which illustrates how early protection can generate readiness that not only secures the gains of conservation but also increases the capacity to address threats elsewhere.[1] In 2000, when access was difficult and pressure on the land was low, residents of Roboré secured protection for 262,305 hectares of forested hills supplying their water (Gobierno Municipal de Roboré 2000, Vera 2016). Work on the San José–Roboré section of the highway to Brazil began in 2003, and as it advanced the reserve became steadily more accessible (IIRSA 2005, Vera 2016). A management committee formed around the reserve, bringing together communities, municipal officials, and conservation partners (Miserendino et al. 2012). This readiness paid off in 2009, when the mining company Kyleno called a public hearing on potential mineral exploration in the area and most communities had little to no information about the proposal. Tucabaca's management committee organized to obtain technical advice, visited the communities, and helped secure a preliminary hearing in Santiago de Chiquitos. By the time the company presented its proposal there, residents were prepared: they voiced a common objection and successfully blocked the proposal (Vera 2016).

[1] The reserve was long known as Tucavaca, the spelling used in some of the sources cited here; the current spelling reflects the name's Chiquitano roots (Requena Gallo 2014).

It paid off again in 2018, when about 55 hectares were cleared in the neighboring municipal reserve of Paquió. Tucabaca's committee joined municipal and civic representatives to inspect the damage and contest further clearing, and following the wider mobilization, Bolivia's land reform agency (INRA) revoked a settlement authorization covering 1,700 hectares (La Región 2018, Agencia de Noticias Fides 2018). Protecting one place had built an institution that later mattered at another.

More broadly, conservation planning is often presented as a question of *where* to act, but, as Tucabaca illustrates, every choice of site also contains a question of *when*, and a single priority ranking collapses two decisions that need not have the same answer. Keeping the siting and timing comparisons distinct reveals whether a parcel is unimportant on current terms, important but not urgent, or important and urgent; this paper separates those decisions and studies how they interact when the act of protecting a parcel also increases readiness.

Separating siting from timing also changes how one reads the evidence on where protected areas are located. The empirical literature warns that protected areas are often located on inexpensive, weakly threatened land with little measured effect on conversion (Joppa & Pfaff 2009, Andam et al. 2008, Pfaff et al. 2009, Börner et al. 2020). That finding identifies a real failure when low cost or political convenience substitutes for ecological contribution, but it does not imply that current threat is itself the conservation objective. Seen this way, the choice between near-and-threatened and remote-and-inexpensive land becomes partly a question of timing: Armsworth (2018) shows that the apparent trade-off depends on how costs and ecological benefits are weighted through time. Remoteness describes both where a parcel is and how long it may remain inexpensive and unthreatened.

On the timing side, the value of waiting for information before an irreversible decision, known as *quasi-option value*, is well established (Arrow & Fisher 1974, Henry 1974, Dixit & Pindyck 1994), but conservation involves irreversibility on both sides of the decision: delay can lose the natural asset or the opportunity to acquire it on favorable terms, while premature action can lock scarce resources into the wrong place (Pindyck 2000, Sims & Finnoff 2016). Work on conservation timing already emphasizes the value of financial flexibility (Yoon & Armsworth 2022, Armsworth et al. 2025). Iacona et al. (2017) show how delaying conservation expenditure can improve outcomes when the resources available for later action grow, using financial returns to illustrate a broader argument that also encompasses learning and capacity building. A further question is

how the timing decision changes when protection itself creates the legal, administrative, and political capacity used in later conservation.

In this paper we allow the agency's capacity to grow through protection, as it did around Tucabaca, so that the act of protection itself changes what the agency can do next, including at other sites. Readiness, the capacity built in this way, can include legal precedent, administrative routines, relationships with regulators and landowners, donor credibility, public legitimacy, and political support, all produced through activity rather than by the passage of time alone. Practitioners have begun to formalize conservation readiness as a static diagnostic of the institutional capacity that makes conservation action feasible and effective (Wollstein et al. 2024, Barr & Lemieux 2021); here, readiness is also produced by protection and enters the decision of when to protect. Where quasi-option value and capacity built while waiting reward delay, readiness is a reason to act earlier.

The paper's main contribution is a dynamic theory of conservation priority that separates siting from timing and adds readiness as an action-generated form of institutional capacity. In a real-options framework, which values the flexibility to delay a costly decision that is hard to reverse, we model a parcel that may be ecologically valuable but is currently far from a development frontier. As the frontier approaches, the opportunity to secure the parcel on favorable terms, whether by purchase, easement, or public designation, can disappear at an uncertain time. Protecting the parcel yields direct conservation benefits and builds readiness for a distinct later conflict at another site, for example a proposal to clear other land the agency would like to protect. Protection thus affects conservation beyond the parcel through three channels, each of which the model formalizes: readiness makes later intervention elsewhere feasible or less costly; protected status can deter development challenges at the parcel before they are made; and development blocked at the parcel can relocate, so the effect on the wider landscape depends on where it goes. Figure 1 summarizes the sequence of events.

The model separates two decisions: whether the benefits of protection, including readiness, exceed its cost under current conditions (its current surplus), and whether now is the best time to protect. As in standard real-options analysis, an acquisition with positive current surplus can still be worth delaying, because delay defers a costly commitment. Both decisions depend on ecological quality (Figure 2(a)). At a lower quality threshold, direct benefits and readiness together cover the cost of acquisition. Immediate protection requires a second, higher threshold,

at which current surplus is also large enough to outweigh the gain from delay. Between the two thresholds lies a genuine waiting region, in which protection is worthwhile but should wait. A more fragile acquisition window lowers the quality threshold for immediate protection. Holding today's price fixed, higher future acquisition prices weakly expand the set of conditions under which immediate protection is optimal. Both mechanisms can support protection before conversion is imminent: development expectations, speculation, and competing uses can close the acquisition window or raise its price early, so a parcel can warrant early protection because the chance to secure it cheaply is disappearing.

Standard real-options reasoning holds that greater uncertainty favors waiting. For readiness, the answer depends on the agency's initial capacity. Uncertainty about the stakes of later conflicts also changes the value of the institutional capacity created by protection: under the model's conditions, greater uncertainty raises the marginal value of readiness – the value of a small addition to capacity – for agencies with low initial capacity and lowers it for agencies with high initial capacity. The positive response is clearest near zero capacity, where a small addition mainly increases access to future intervention, and greater uncertainty makes that opportunity more valuable. A change in readiness value does not by itself settle the timing decision, because uncertainty also changes the value of waiting to protect. For an agency with sufficiently low initial capacity, however, a sufficiently fragile acquisition window limits the attraction of delay: greater uncertainty then lowers the ecological quality needed to justify immediate protection, bringing protection forward.

The framework also bears on how measures of conservation impact are used in decisions. We use *outcome additionality* to mean the causal difference between outcomes with and without protection over a stated horizon, distinct from the project- or financial-additionality tests used by some funding mechanisms. Additionality thus measures what protection changed over the given horizon, which makes it central to evaluating conservation policy. As a criterion for deciding where and when to protect, however, it is a reliable guide only when it agrees with the agency on which parcel is preferable at a common date and on when a given parcel should be protected. Each condition can fail: a threat that arrives after the evaluation horizon can reverse the ranking of parcels, and an eligibility threshold can delay protection until favorable acquisition terms are gone. A site-level evaluation also leaves out effects beyond the protected site, such as the readiness gained for later conflicts elsewhere and the damage done by displaced development, and

evidence based on records of observed challenges (documented attempts to overturn protection) cannot show the challenges that protection deterred. Even for avoided-deforestation credits, where additionality defines the service being credited (Groom & Venmans 2023), the value of acquiring the land can fall as clearing risk rises, if acquisition costs rise faster than the credited benefit. The domain of additionality should match the decision it is asked to make.

Section 2 develops the conceptual framework. Sections 3 and 4 present the model and the timing results. Section 5 examines how readiness affects contests over protected land and how targeting shapes the landscape effects of displaced development. Section 6 considers the implications for additionality-based selection, funding, and evaluation. Section 7 discusses conservation practice, finance, and the model's limitations. The appendix gives notation and proof overviews; an online supplementary appendix[2] gives the derivations and proofs, robustness results, the valuation of the later intervention, the strategic extension, and figure notes.

2 Conservation priority over space and time

2.1 Where to act and when to act

Siting priority compares parcels at a common date; *timing priority* compares protection dates for a given parcel. Both comparisons weigh expected net conservation benefits over the agency's horizon. These may include direct ecological benefits, complementarity with the existing portfolio, acquisition and management costs, the risk that the opportunity disappears, and the effect of protection on future conservation capacity. A full dynamic portfolio problem combines the two comparisons; keeping them distinct helps explain why a particular parcel or acquisition date is preferred. Table 1 summarizes the two comparisons, alongside a third question, what protection changed, that Section 6 takes up.

Existing approaches differ in which of these comparisons they emphasize. Systematic conservation planning is principally organized around the siting comparison, asking how a set of sites can meet ecological objectives given costs, representation targets, complementarity, and other constraints (Margules & Pressey 2000). Threat-weighted return-on-investment rules add urgency to that spatial comparison, and work well when threat is a good proxy for the risk of

[2]Available at https://dmweinhold.github.io/Latest-Version-Papers/Weinhold_Readiness_online_supplement.pdf.

Table 1: Conservation siting, timing, and evaluation

Object	Central question	Principal inputs	Natural horizon
Siting priority	Which parcel contributes more if action occurs at a common date?	Ecological contribution, representation, complementarity, connectivity, condition, permanence, cost, and cross-site effects	Agency's portfolio horizon
Timing priority	For a given parcel, should the agency act now or wait?	Threat path, acquisition-price path, availability, information, finance, irreversibility, and institutional state	Decision and option horizon
Outcome additionality	What outcome changed because protection occurred?	Counterfactual loss, protection effectiveness, measured outcome, and evaluation window	Evaluation or crediting horizon

losing valuable options and when costs and effectiveness are adequately measured. The empirical location-bias literature appropriately warns that choosing land merely because it is inexpensive or politically easy can generate protected areas with little causal effect on land conversion (Joppa & Pfaff 2009, Andam et al. 2008, Pfaff et al. 2009, Börner et al. 2020).

Dynamic reserve-selection models make the timing comparison explicit: unprotected sites can disappear, prices and budgets evolve, and apparently attractive plans can be overtaken by events (Meir et al. 2004, Costello & Polasky 2004, Strange et al. 2006, McDonald-Madden et al. 2008). Pre-emptive conservation and "fire-fighting" can then be compared as alternative policies for an evolving set of opportunities (Spring et al. 2007). The framework of Armsworth (2018) is closest to the distinction drawn here, since distance from a development frontier is a matter of time as well as space; Armsworth et al. (2025) generalize it by letting conservation return vary through time with biodiversity status, threat, cost, funding, and parcel availability.

Real-options analysis shows that timing depends on more than urgency, because the value of waiting has to be weighed against the value of acting early (Arrow & Fisher 1974, Henry 1974, Dixit & Pindyck 1994, Pindyck 2000). Financial instruments can partly separate the two: borrowing, revolving funds, multiyear budgets, and rights of first refusal let agencies secure valuable sites with a partial or deferred commitment (Yoon & Armsworth 2022, Armsworth et al. 2025). The agency's task is to identify the threshold, which depends on current conditions, at which the value of acting overtakes the value of waiting.

2.2 Readiness as capacity that protection produces

Readiness is the institutional capacity created through protecting and managing land. Its usefulness depends on what experience produces and whether those capabilities transfer to later conservation problems. Organizational economics provides a basis for treating such capacity as the accumulated product of activity: learning-by-doing, organizational-learning, absorptive-capacity, and dynamic-capabilities theories all emphasize that prior related experience improves the ability to recognize opportunities, coordinate action, and reconfigure resources (Arrow 1962, Cyert & March 1963, Nelson & Winter 1982, Levitt & March 1988, Cohen & Levinthal 1990, Teece et al. 1997, Zollo & Winter 2002). Conservation capacity-development research similarly emphasizes learning during implementation and the interaction between individual, organizational, and societal capacity (Porzecanski et al. 2022). Conservation evidence links institutional and management capacity to stronger protected-area performance (Bonilla-Mejía & Higuera-Mendieta 2019, Powlen et al. 2021).

In conservation, this internal capacity can be highly specific. Protecting a parcel in one jurisdiction may establish legal precedent, staff routines, regulator relationships, donor confidence, and monitoring systems that are useful in a later conflict at another site. The value depends on relevance: experience in the same biome, legal regime, agency network, or political coalition should transfer more strongly than unrelated activity. The model captures this connection with a relevance parameter rather than assuming that every protected hectare builds equivalent capacity.

Protection can also create external readiness. Once a landscape is protected and used, the experience of living with its protected status can foster familiarity, attachment, and public legitimacy, and the landscape can become a reference point for residents, visitors, donors, agencies, and non-use beneficiaries. Policy feedback can create constituencies that defend existing benefits against retrenchment (Pierson 1993, 1994, Campbell 2003); environmental valuation and place-attachment research provide parallel reasons why protected status can become politically durable (Weisbrod 1964, Krutilla 1967, Williams & Vaske 2003). This external readiness also affects fully rational developers, who anticipate higher legal, political, reputational, or administrative costs of challenging protection once it has built a defensible status quo (Section 5).

Readiness differs from conventional learning by doing, in which experience lowers the cost of repeating the same task (Arrow 1962), in two ways. First, readiness acts on the ability to intervene as well as on the cost of intervention. Second, part of it resides outside the agency, in legal precedent, political constituencies, donor confidence, and public legitimacy (Pierson 1993, Campbell 2003). The model adds this stock of organizational and political capital to a conservation real-options problem, so that the case for acting early can be weighed directly against quasi-option value.

Readiness therefore changes the state in which later siting and timing decisions are made. It also gives a reason why some early protections have system value while others do not: readiness must be produced, relevant to later conflicts, and persistent enough to outweigh the direct cost and the value of waiting.

3 Model

3.1 Overview

A broader conservation-planning exercise has identified a candidate parcel, R, whose ecological contribution may be high. Here *acquisition* means any action that secures protection, whether purchase, a conservation easement, or public designation. The model evaluates the value of acquiring it now and the gain from waiting. These two comparisons are the formal counterpart of the distinction between siting priority and timing priority drawn in Section 2. Both depend on its marginal contribution to the current portfolio and the opportunity cost of scarce conservation resources. The central timing question addressed here is how the capacity created by protection changes the case for early acquisition.

Parcel R is currently far from the development frontier. It is inexpensive to protect in both financial and political terms, but its near-horizon probability of conversion is low. The frontier is uncertain, however. As it approaches, landowners' outside options rise, developers take interest, anticipated rents capitalize into land prices, and a routine acquisition can become more expensive, economically, politically, or both. For example, in frontier settlements of the western Brazilian Amazon, distance to market alone explains close to one-third of the variation in farm land values, consistent with a von Thünen gradient, in which land values fall with distance from markets (Sills & Caviglia-Harris 2008), and structural estimates confirm that

lower transport costs raise the private demand for clearing (Souza-Rodrigues 2019). A parcel that is currently cheap *because* it is remote can therefore become expensive, and competed for, through anticipated access, well before loss is imminent. We call this period of favorable terms the *acquisition window* and represent its closure by a *hazard rate* γ_R, the probability per unit of time that the window closes, given that it is still open.

Protecting R also changes the agency. It adds readiness to the agency's institutional capacity S, a stock of legal precedent, administrative experience, public legitimacy, and political goodwill. That capacity is useful when a *later conflict*, C, arises at another site: a developer targets land the agency would like to conserve, the conservation stakes X_t are uncertain, and the agency decides whether and when to intervene. Greater capacity makes it more likely that the agency can intervene at all and, when intervention is feasible, reduces its cost. Protecting R therefore has value both for the direct conservation benefit it secures and for the readiness it creates for later conflicts. At Tucabaca, the agency in this sense was the municipality of Roboré together with the committee formed around the reserve, and the clearing at Paquió in 2018 was a later conflict of this kind.

The model abstracts from a full reserve-network problem and from the spatial evolution of the development frontier, including adjacency, leapfrogging, and the redirection of development. It carries one implication of spatial dynamics: a currently favorable acquisition window can close. Section 5 extends it to the wider landscape.

Once R is protected, a developer may challenge its protected status, for example by seeking to have the protection downgraded or removed; we call such a challenge a *contest*. The 2009 mining proposal inside Tucabaca is an example. Readiness at R can deter contests before they are made. Deterred contests are not observed, but they have two concrete effects: R stays protected with no contest on record, and a developer who is deterred, like one who is defeated in a contest, may relocate the project. Section 5 examines what protection achieves across the landscape when blocked development moves elsewhere, and how the choice of protected site shapes that outcome.

Figure 1 shows the sequence of events. The agency first decides when to protect R. Until it does, the acquisition window on R may close, or the later conflict C may arrive at another site. Protecting R adds readiness to the agency's capacity, and that readiness may depreciate before C arrives. When C arrives, the agency's capacity determines whether it can intervene

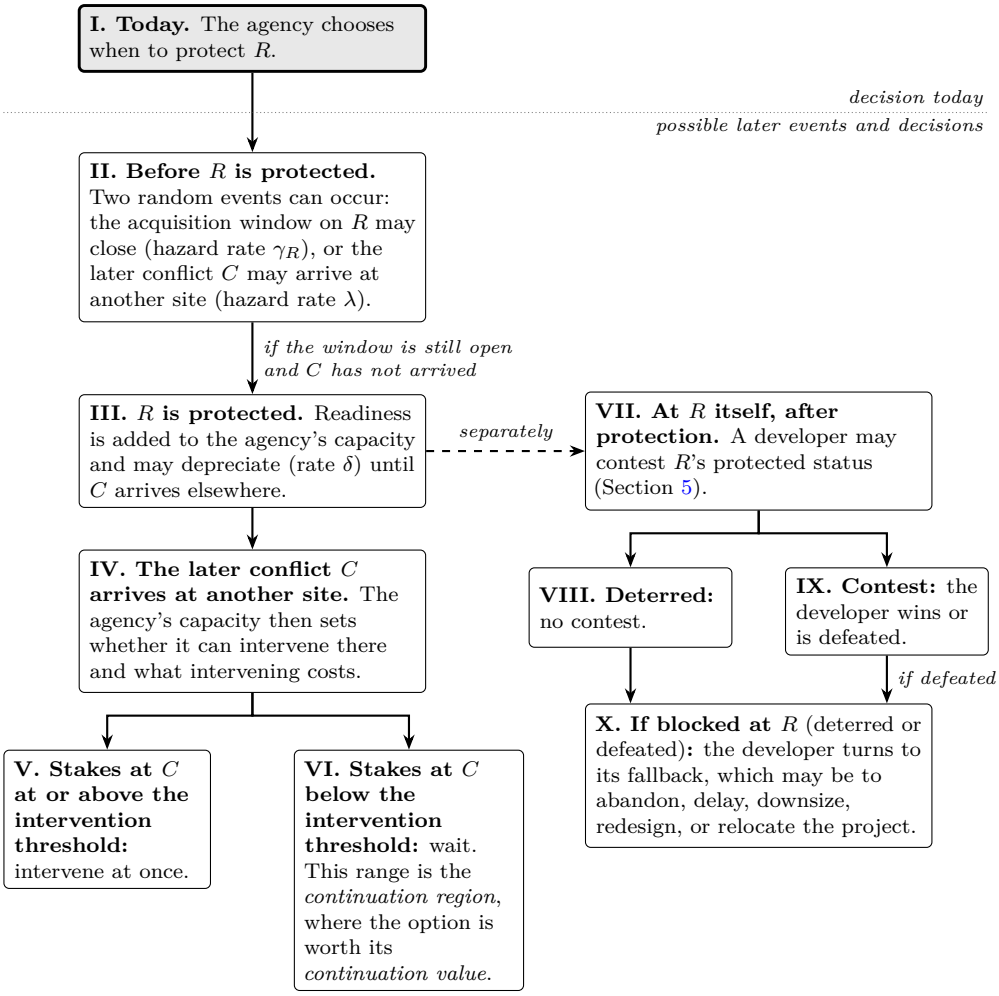


Figure 1: **Sequence of events.** Box I, shaded, is the decision the agency makes today; boxes II–X are possible later events and decisions. The left column shows the agency's problem (Sections 3 and 4); the right column shows a developer's response at the protected parcel R (Section 5). A contest is a developer's attempt to overturn protection already in place at R; the later conflict C arises at another site, where the agency decides whether to intervene. Boxes V and VI separate stakes at C that already justify intervening from stakes that do not. Following the real-options literature, we call the dividing level the *intervention threshold*, the range below it the *continuation region*, and the value of the agency's right to intervene later, while it waits, its *continuation value*. A hazard rate is the probability per unit of time that an event occurs, given that it has not yet occurred. The readiness-dependent acquisition path requires protection before either event in box II. If direct benefits alone cover acquisition cost, immediate protection is already optimal.

there and at what cost; it intervenes at once if the stakes at C are already high enough, and otherwise waits until they are. Separately, once R is protected, a developer may contest R's protected status or be deterred from doing so, and a developer blocked at R turns to its *fallback*: abandoning, delaying, downsizing, redesigning, or relocating the project. The rest of this section sets out the agency's problem; Section 5 returns to the developer.

3.2 The candidate parcel and its acquisition window

The model evaluates one candidate conditional on the agency's existing conservation portfolio and resource constraints. Ecological quality is fixed at $q_R \geq 0$, and, conditional on that portfolio, securing R yields direct benefit $B_R(q_R)$ over the agency's horizon, the marginal contribution to the portfolio that protection secures, with $B_R(0) = 0$ and $B_R' > 0$. The direct benefit can reflect representation, connectivity, complementarity, and redundancy. Current remoteness implies low near-horizon conversion pressure, but it does not determine q_R. Readiness is likewise valued at the margin: experience that duplicates existing capabilities may contribute less than experience that fills an institutional gap.

The current effective acquisition cost, $P_R > 0$, is interpreted broadly to include monetary cost, political effort, and the opportunity cost of the conservation resources that protection uses, with each cost counted only once.[3] Throughout the focal timing problem, the surrounding portfolio and the opportunity cost of resources are held fixed. Time is continuous, $t \in [0, \infty)$, and the agency discounts future benefits and costs at rate $r > 0$. The discount rate is the opportunity cost of scarce money, staff, legal effort, and political capital. It is not a claim that future biodiversity is intrinsically less valuable. Direct net value is

$$D_R(q_R) = B_R(q_R) - P_R,$$

and, in the core region where direct net value is negative, the direct shortfall is

$$L_R(q_R) = P_R - B_R(q_R) > 0.$$

The acquisition window closes at a random time $T_R \sim \text{Exp}(\gamma_R)$, with $\gamma_R > 0$. The assumption $\gamma_R > 0$ means that favorable acquisition terms are not guaranteed to last indefinitely, although the hazard rate may be arbitrarily small. Appendix A collects the core notation.

[3] For example, write $P_R = d_R + \nu c_R$, where d_R is a direct burden not already represented by use of a common resource, c_R is the amount of that resource used, and ν is its marginal opportunity cost. In a continuous portfolio problem, ν is the multiplier on the resource constraint. More generally, it represents the value of displaced conservation opportunities.

3.3 Institutional readiness and the later conflict

The agency's institutional capacity is a stock $S_t \geq 0$. Protecting R adds readiness $\kappa_R(q_R, \rho_R)$ to it, where $\rho_R \geq 0$ measures how relevant the protection is to later conflicts, for example when both involve the same legal regime. We assume

$$\kappa_R(0, \rho_R) = \kappa_R(q_R, 0) = 0, \qquad \frac{\partial \kappa_R}{\partial q_R} > 0 \text{ for } \rho_R > 0, \qquad \frac{\partial \kappa_R}{\partial \rho_R} > 0 \text{ for } q_R > 0.$$

Higher ecological quality and greater institutional relevance are assumed to increase the readiness created by protection, for example because more valuable sites attract stronger constituencies and set more consequential precedents. The model studies what follows when protection produces capacity that is useful in later conflicts; protection with no relevance to them produces none. We represent these gains as an initial addition to the agency's capacity. This simplification captures the institutional consequences of establishing protection without separately modelling the gradual development of public familiarity or political support after designation.

The new capacity may lose some of its usefulness before a later conflict arises, so we allow the readiness increment to depreciate at rate $\delta \geq 0$, leaving $\kappa_R(q_R, \rho_R)e^{-\delta t}$ available t periods after protection. The agency's pre-existing capacity S is maintained at its current level, so depreciation applies only to the additional capacity created by protecting R.

The stakes at C, the conservation value at stake there, follow a geometric Brownian motion: they grow on average at rate μ and fluctuate randomly, with volatility σ measuring the size of the fluctuations,

$$dX_t = \mu X_t \, dt + \sigma X_t \, dW_t, \qquad r > \mu, \quad X_0 = x > 0, \quad \sigma > 0,$$

and the conflict arrives at $T_C \sim \text{Exp}(\lambda)$, with $\lambda > 0$, independent of T_R. The stakes X_t evolve independently of both event times. The two hazard rates have different meanings: γ_R governs the closing of the current acquisition window, while λ governs the arrival of the later conflict in which readiness is useful. The agency observes the stakes before the conflict arrives. The conservation value of land elsewhere that a later proposal could threaten can be assessed before a development proposal, permit process, or legal dispute opens the opportunity to intervene; the opening of that opportunity is the separate random event that arrives at rate λ.

Throughout, capacity available when conflict C arrives determines the agency's access and intervention-cost terms for that conflict. This reflects capacity that must be in place before a proceeding opens, such as legal standing, recognized relationships with regulators, or administrative capability. Those terms remain fixed while the agency decides when to intervene, so acquiring R after the conflict has arrived cannot improve them; readiness has to be built beforehand.

When the later conflict C arrives at another site, an agency that is able to intervene does not have to act at once. It can pay the intervention cost $I(S)$ at any time, and waiting has value because the stakes are uncertain and the agency keeps the choice of whether to intervene. Following convention in the real-options literature, we call this right to intervene later, which does not expire, an *option* (Dixit & Pindyck 1994, Chapter 5). Arrival of the conflict opens the opportunity to intervene; treating that opportunity as open-ended is a simplification that abstracts from deadlines a proceeding may impose. The agency intervenes once the stakes reach the *intervention threshold* $x^*(S)$, the level at which intervening becomes better than waiting longer. Below that threshold, in what is called the *continuation region*, the agency waits, and the value of holding its option is its *continuation value* (box VI of Figure 1). At or above the threshold, it intervenes at once and receives the net payoff from intervening (box V). Online Appendix S.1 gives the option value $F(x; I)$, which combines the two cases, and the intervention threshold $x^*(S)$.

Readiness raises the probability that intervention is feasible, represented by $p(S) \in [0, 1]$ with $p'(S) > 0$, and lowers the intervention cost, $I'(S) < 0$. Accounting for access, the value of the later intervention opportunity is

$$\mathcal{F}(x, S) = p(S)F(x; I(S)).$$

The access probability $p(S)$ for the later conflict C at another site, together with other political and constituency effects, is represented in reduced form. Section 5 studies an analogous external-readiness mechanism at the focal parcel R, where stronger protected status can deter contests before they are made.

Protecting R when the stakes are x creates the *readiness value*

$$K(x, S, \kappa) = \mathbb{E}_x\left[e^{-rT_C}\{\mathcal{F}(X_{T_C}, S + \kappa e^{-\delta T_C}) - \mathcal{F}(X_{T_C}, S)\}\right], \qquad \kappa = \kappa_R(q_R, \rho_R). \tag{1}$$

This is the value today of the capacity that protection adds, measured by how much it improves the agency's position when the later conflict arrives. It accounts for both cases at arrival, a conflict whose stakes already justify intervening (box V) and one in which the agency waits (box VI); for the share of the new capacity that has depreciated by then; and for discounting. It allows a readiness increment of any size. Readiness value rises with the stakes at C, and it does not depend on the hazard rate γ_R: the closing of the acquisition window ends the opportunity to protect R, but not the benefits of readiness once R is protected.

4 Value and timing results

With readiness value K included alongside the parcel's direct conservation benefit, we now ask when the agency should protect R. The agency observes the stakes at C and chooses when to protect R in response to them (box I of Figure 1). This describes an agency that monitors conditions elsewhere and can act when they change, for example with bridge finance or a revolving fund. Protection goes ahead at the chosen date provided the acquisition window on R is still open and, when readiness is needed to justify protection, the later conflict C has not yet arrived. Online Appendix S.6 compares a version in which the agency must fix the protection date in advance.

We answer that question in four steps. Section 4.1 separates whether protection is worth its cost from whether it is worth doing now, and shows that ecological quality sorts parcels into the three groups of Figure 2(a). Section 4.2 shows what moves the boundaries between those groups, including a more fragile acquisition window and rising acquisition prices, either of which can support protection before conversion is imminent. Section 4.3 shows how greater uncertainty about later conflicts changes the marginal value of readiness, and that the direction depends on the agency's initial capacity (Figure 2(b)). Section 4.4 asks when that change in value also changes the decision to protect, since uncertainty also changes the value of waiting.

4.1 When protection is valuable, and when to act

We first ask whether protecting R is worth its cost under current conditions. *Current surplus* is the benefit of protection, including readiness, less its cost: $\mathrm{SV}_R(q_R) = B_R(q_R) - P_R + K(x, S, \kappa_R(q_R, \rho_R))$. Let q^0 denote the quality at which this surplus is zero. If direct benefit

and readiness production rise with quality, $B'_R > 0$ and $\partial\kappa_R/\partial q_R \geq 0$, and their combined value eventually exceeds P_R, the threshold exists and is unique. Parcels at or above q^0 have non-negative current surplus, conditional on the existing portfolio and resource costs. Remoteness alone does not help a parcel pass this test, which marks the boundary between the first two zones of Figure 2(a). A currently remote parcel can pass it, but low price alone does not make a parcel worthwhile: inexpensive land of low ecological quality can still fail.

Positive current surplus does not settle when to protect. Suppose direct benefit falls short of acquisition cost by the direct shortfall $L_R(q_R) = P_R - B_R(q_R) > 0$, the part of the cost that direct benefits do not cover. This is the paper's core case, because readiness, rather than the parcel's direct value, is then the reason to protect it before the acquisition window closes. The agency protects R only if both events in box II of Figure 1 remain in the future, since once the later conflict has arrived, protecting R can no longer build readiness for it. If direct benefit instead covers acquisition cost, R remains worth protecting for its direct value even if the conflict arrives first, and immediate protection is optimal.

Waiting therefore has a cost and a benefit when readiness is the reason to protect. It risks losing readiness if the window on R closes or the conflict arrives first, which happens at the combined hazard rate $\gamma_R + \lambda$. It also lets the agency retain the choice to protect R later, if the observed stakes at C rise and the readiness it would create becomes valuable enough to justify immediate acquisition.

Readiness value K measures what readiness adds to the value of protecting now. The decision to protect immediately uses the *readiness contribution to immediate protection*, Q, which enters after the opportunity to wait has been taken into account and is smaller than K when readiness is positive:

$$(2) \qquad Q(x, S, \kappa) = K(x, S, \kappa) - \frac{xK_x(x, S, \kappa)}{\alpha}, \qquad \tfrac{1}{2}\sigma^2\alpha(\alpha - 1) + \mu\alpha = r + \lambda + \gamma_R, \quad \alpha > 1.$$

The root α summarizes how quickly the opportunity to protect R can be lost, through the closing of the window or the arrival of the conflict, relative to how the stakes evolve.

Proposition 1 (When to protect)**.** *Suppose the agency observes the stakes at C and can protect R at any time before the acquisition window closes or the later conflict arrives. If direct benefit covers acquisition cost, immediate protection is optimal. If direct benefit falls short of acquisition*

cost and $\kappa_R(q_R, \rho_R) > 0$*, the agency protects* R *as soon as the stakes reach a unique finite threshold. Immediate protection is optimal if and only if*

$$B_R(q_R) + Q(x, S, \kappa_R(q_R, \rho_R)) \geq P_R. \tag{3}$$

For a positive readiness increment and $x > 0$*, the readiness contribution satisfies* $0 < Q < K$*. The result allows any finite positive readiness increment and any depreciation rate* $\delta \geq 0$*. If* $\kappa_R(q_R, \rho_R) = 0$*, then* $K = Q = 0$*: protection is never undertaken when* $B_R(q_R) < P_R$*, while immediate protection is optimal when* $B_R(q_R) \geq P_R$*.*

The two tests can be compared directly:

Question	Criterion
Does protection have non-negative current surplus?	$B_R + K \geq P_R$
Is immediate protection optimal?	$B_R + Q \geq P_R$

For positive readiness, $Q < K$, so the second test is more demanding. Readiness must do more than cover the direct shortfall: protecting now also gives up the opportunity to postpone that shortfall and to wait for conditions in which readiness is worth more. The acquisition rule accounts for this opportunity by requiring the smaller amount Q, rather than the full readiness value K, to cover the shortfall. Online Appendix S.2 gives the proof.

For a given parcel, ecological quality is fixed, and the agency responds to the evolving stakes at C; the two quality thresholds compare candidate parcels under the same observed conditions, holding acquisition cost and the other inputs fixed. Let q^* denote the quality at which $B_R(q^*) + Q(x, S, \kappa_R(q^*, \rho_R)) = P_R$. Under the same monotonicity assumptions, q^* is unique when it exists. If the current-surplus threshold q^0 creates positive readiness, then $q^* > q^0$ because $Q < K$: between the two thresholds, current surplus is positive but immediate protection is not optimal, the middle zone of Figure 2(a). The middle zone is economically important because it separates siting priority from timing urgency: the parcel is worth securing, but not yet. Higher ecological quality brings protection forward, because it shrinks the direct shortfall and, if higher-quality protection builds more readiness, raises Q.

Figure 2(a) shows the three groups. The first is an assessment under current conditions. When protection would create positive readiness, the opportunity to protect R later still has

positive value, because the stakes at C may rise far enough to justify acquisition (Proposition 1). Negative current surplus from protecting now is therefore not a rule for excluding a parcel permanently; that would require a broader portfolio decision or additional frictions.

4.2 What brings protection forward?

The rule in Proposition 1 shows what moves the boundaries between the groups in Figure 2(a). The risk that favorable terms will disappear is a reason to act sooner, but it does not by itself give a parcel positive current surplus. A more fragile acquisition window, a higher hazard rate γ_R, leaves readiness value K unchanged, since the window affects only the opportunity to protect R. It raises Q, because waiting is then a worse gamble, and so lowers q^* and compresses the waiting region. Holding other conditions fixed, a less fragile acquisition window makes waiting a safer bet and widens the waiting region.

The value of readiness and the cost of protection move the thresholds as well. More productive or more relevant readiness raises the readiness increment κ_R, which raises both K and Q and lowers both thresholds. Faster depreciation of readiness lowers Q and raises q^*, and a higher current acquisition cost raises both thresholds. A later conflict expected sooner, a higher λ, raises readiness value, because readiness is worth more when it is likely to be used sooner. It also lowers the quality threshold for immediate protection when readiness does not depreciate. For a small addition to readiness, the same timing result holds to first order when the window-closure rate is at least the depreciation rate, $\gamma_R \geq \delta$ (Online Appendix S.2). The arrows in Figure 2(a) show the main shifts.

Proposition 1 holds the acquisition cost fixed, which is conservative for the case for early protection. Rising acquisition prices strengthen that case.

Proposition 2 (Higher future acquisition prices)**.** *Hold today's acquisition price fixed, and suppose every future acquisition price is at least as high as before. Then the value of the opportunity to protect R cannot rise, and if immediate protection was optimal before, it remains optimal.*

Higher future prices lower the payoff from every policy that protects later, while leaving immediate protection unchanged, so the set of conditions under which immediate protection is optimal can only expand. The comparison allows prices to follow any random path and does not

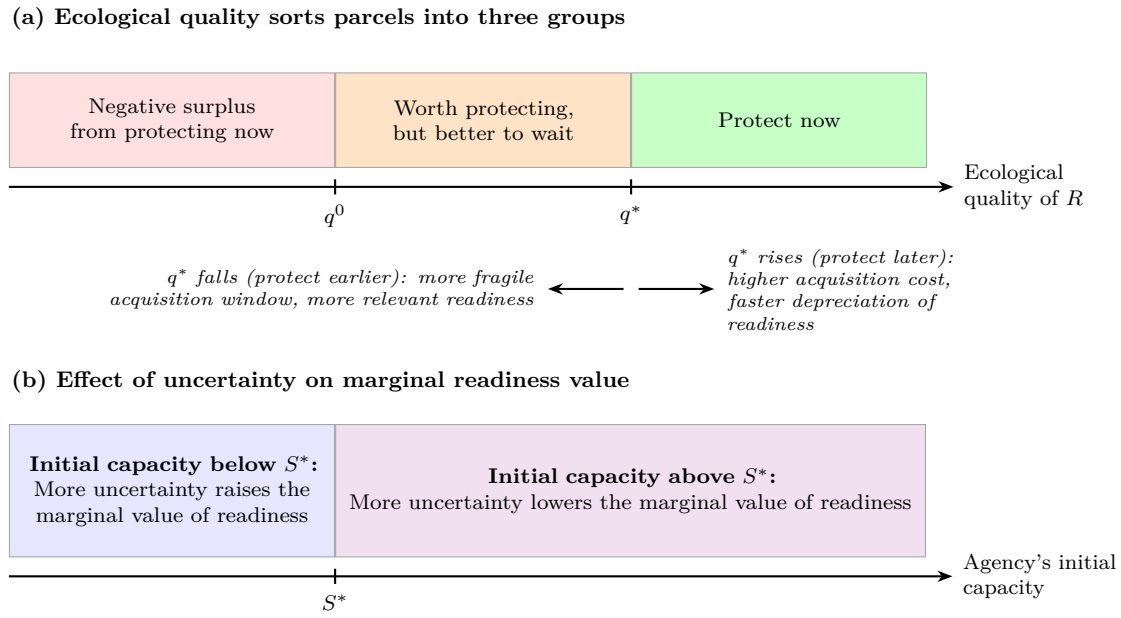


Figure 2: **Results at a glance.** Panel (a): ecological quality sorts candidate parcels into three groups (Proposition 1). Below q^0 the current surplus from protecting now is negative, so protection is not worthwhile on current terms, although it may become worthwhile later; between q^0 and q^* protection is worthwhile but waiting is better; at or above q^* the agency should protect now. The arrows show what moves q^* (Section 4.2). Holding today's acquisition price fixed, higher future prices weakly expand the set of conditions under which immediate protection is optimal (Proposition 2). Panel (a) compares ecological quality while holding current acquisition cost, observed stakes, initial capacity, and institutional relevance fixed, with positive readiness at the current-surplus threshold. Panel (b): greater uncertainty about the stakes at C raises the marginal value of readiness when the agency's initial capacity is below S^*, and lowers it above S^* (Proposition 3). Whether that change also brings protection forward depends on the value of waiting (Section 4.4). Both panels are schematic; Online Appendix S.8 gives the underlying conditions.

require a constant acquisition threshold (Online Appendix S.3.2). This gives the price channel a direct timing role: waiting can preserve flexibility, but it can also sacrifice the chance to protect R cheaply before ecological conversion becomes imminent. Effective cost can also rise when the agency's scarce resources become more valuable elsewhere; how that value evolves is a question for a full portfolio model.

4.3 Uncertainty and the marginal value of readiness

A standard intuition from real-options analysis is that greater uncertainty favors waiting. Uncertainty about the stakes at C, measured by their volatility σ, matters here in two ways: it changes the value of the readiness that protection creates, and it changes the value of waiting to protect. This subsection considers the first; Section 4.4 considers the second.

To isolate how readiness value responds, consider the marginal value of capacity at the agency's current level,

$$\Omega(x,S) = \left.\frac{\partial K(x,S,\kappa)}{\partial \kappa}\right|_{\kappa=0} = \int_0^\infty \lambda e^{-(r+\lambda+\delta)t}\,\mathbb{E}_x[\mathcal{F}_S(X_t,S)]\,dt. \tag{4}$$

This is the value of a little more capacity, including states in which the later conflict arrives with stakes that already justify intervening (box V). The derivative inside this expectation separates the two gains from a small addition to capacity:

$$\mathcal{F}_S(X_t,S) = \underbrace{p'(S)F(X_t;I(S))}_{\text{access gain}} + \underbrace{p(S)I'(S)F_I(X_t;I(S))}_{\text{cost saving}}. \tag{5}$$

The first term is the gain from making intervention feasible; the second is the saving when intervention is already feasible, the part closest to conventional learning by doing, in which prior activity lowers the cost of a later intervention. The marginal readiness value Ω averages these gains over possible conflict-arrival dates and stakes, accounting for depreciation and discounting.

The response of Ω to uncertainty reflects both access and cost savings. Greater uncertainty makes the access benefit more valuable. Its effect on the cost-saving benefit depends on the stakes relative to intervention cost: that benefit can become more or less valuable. Under the following conditions, the combined response is positive at low initial capacity and negative at high initial capacity, with a single crossing.

Proposition 3 (Uncertainty and the value of readiness)**.** *Suppose that access is zero for an agency with no capacity and, at every capacity level, increases at a diminishing rate, while intervention cost falls at a constant or increasing proportional rate:*

$$p(0)=0, \quad p'>0, \quad p''\le 0, \quad 0\le p\le 1, \qquad \chi_I(S) = -I'(S)/I(S) > 0, \quad \chi_I'(S)\ge 0.$$

Fix the current stakes $x>0$ and volatility $\sigma>0$. There is a unique capacity threshold $S^*>0$ such that greater uncertainty raises the marginal value of readiness, $\Omega_\sigma>0$, for $S<S^*$, and lowers it, $\Omega_\sigma<0$, for $S>S^*$.

At very low capacity, a small addition to capacity mainly creates access to the later intervention, and uncertainty makes that opportunity more valuable. As capacity rises, further access gains diminish. At sufficiently high capacity, greater uncertainty reduces the cost-saving

component of marginal readiness value by more than it increases the access component. Readiness value measures an improvement over the agency's existing intervention opportunity, and uncertainty can make both more valuable while shrinking the gain from adding capacity. Under the stated conditions, the combined response changes sign once. Figure 2(b) shows the two zones. The result holds at any level of today's stakes. The exponential access and cost functions $p(S) = 1 - e^{-\psi S}$ and $I(S) = I_0 e^{-\chi S}$ satisfy the conditions, and Online Appendix S.4 provides the proof.

4.4 When uncertainty changes the decision to protect

More valuable readiness need not mean more urgent protection: uncertainty also changes the attraction of waiting. Proposition 1 gives the exact condition. Greater uncertainty lowers the quality threshold for immediate protection if and only if it raises the readiness contribution to immediate protection, Q:

$$\frac{\partial q^*}{\partial \sigma} = -\frac{Q_\sigma}{B_R'(q^*) + Q_\kappa \, \partial \kappa_R / \partial q_R}, \tag{6}$$

where the denominator is positive. The response Q_σ combines the change in readiness value, the change in its sensitivity to the stakes, and a negative term from the acquisition root α. This last term works against immediate protection. As the acquisition window becomes more fragile, the adjustment between Q and K shrinks, allowing a strictly positive readiness-value response to determine the timing response under the conditions below.

Proposition 4 (Uncertainty when the acquisition window is fragile)**.** *Suppose the current-surplus threshold q^0 is interior, protection there creates positive readiness, and greater uncertainty raises the readiness value created by protection at q^0, $K_\sigma > 0$. Then there is a finite window-closure rate $\bar{\gamma}_R$ such that, for every $\gamma_R > \bar{\gamma}_R$, greater uncertainty lowers the quality threshold for immediate protection, $\partial q^*/\partial \sigma < 0$.*

Corollary 1 (Low initial capacity)**.** *Suppose $p(0) = 0$, and that at zero capacity the current-surplus threshold is interior and creates a positive readiness increment. Then there are a capacity level $\bar{S} > 0$ and a finite window-closure rate $\bar{\gamma}_R$ such that greater uncertainty lowers the quality threshold for immediate protection whenever $S \leq \bar{S}$ and $\gamma_R > \bar{\gamma}_R$. The result allows any finite readiness increment and any depreciation rate.*

At zero capacity the agency has no access to the later intervention, so the capacity created by protection supplies access to an option whose value rises with uncertainty, and there is no existing option value to subtract. For an agency with sufficiently low initial capacity facing a sufficiently fragile acquisition window, greater uncertainty about later conflicts therefore lowers the quality needed for immediate protection, bringing protection forward against the usual presumption that uncertainty is a reason to wait.

How fragile the window must be depends on conditions. In the calculations of Online Appendix S.5, greater uncertainty already lowers the level of the stakes at which a low-capacity agency protects R, and raises it for a capable agency, under the parameterization reported there. Here, bringing protection forward means that immediate protection becomes optimal under a wider set of current conditions.

5 Deterrence, displacement, and the wider landscape

The timing model values the capacity that protecting R creates for intervention at another site, C. Protection can also build a more defensible status quo at R itself. Taking protection of R as given, this section examines that separate strategic implication: how institutional capacity changes a developer's decision to challenge protected status, and how the resulting deterrence and displacement – contests that never happen and development that moves elsewhere – affect ecological outcomes across the landscape.

5.1 Readiness, contest formation, and observed outcomes

A contest is a developer's attempt to overturn protection already in place at R, for example by seeking to have it downgraded or removed (box VII of Figure 1). Such attempts are common enough to have a name in the conservation literature, protected area downgrading, downsizing, and degazettement (Mascia & Pailler 2011, Golden Kroner et al. 2019). Readiness can help defend R by making protected status more costly to contest. A developer contests protection only when the expected gain exceeds the cost of contesting, so greater conservation capacity can prevent a contest from forming. This is a strategic consequence of readiness at R, distinct from the access probability $p(S)$ for intervention in the later conflict C at another site. Online Appendix S.7 supplies a simple developer-threshold model.

The accounting must separate three things protection can do: defeat some contests that form, deter others from forming, and redirect blocked development elsewhere. Let $c(S)$ be the probability of a contest, holding developer strength fixed, and suppose it decreases with conservation capacity. Let ξ be the probability that R would be developed without protection, $\Lambda_R > 0$ the resulting ecological loss, and $\pi_D \in (0,1)$ the developer's probability of winning a contest. Assume $0 \leq c(S) \leq \xi$: contests come from projects that would have developed the parcel if it were unprotected. Holding ξ and π_D fixed isolates the effect of readiness on contest formation.

Only a contest that the developer wins produces focal loss under protection. The essential accounting is therefore

$$
\begin{aligned}
A_R^{\text{obs}} &= (1-\pi_D)c(S)\Lambda_R, \quad \text{successful defenses in observed contests,} \\
A_R^{\text{di}} &= [\xi - \pi_D c(S)]\Lambda_R, \quad \text{total focal avoided loss.}
\end{aligned} \tag{7}
$$

Their difference, $[\xi - c(S)]\Lambda_R$, is loss avoided through deterrence. An observation of "no contest" cannot distinguish a parcel that was never at risk from a contest that protection deterred. A credible comparison of outcomes with and without protection can, in principle, recover the full focal effect.

Proposition 5 (The observed-contest gap)**.** *With counterfactual development risk and conditional defense effectiveness fixed, greater readiness that reduces contest formation lowers contest-visible avoided loss while raising total focal avoided loss. Under full deterrence, the contest-visible measure is zero even though total focal avoided loss is positive whenever* $\xi > 0$.

The result follows directly from equation (7). Of the contests that additional readiness newly deters, a fraction $1-\pi_D$ would already have been defeated. Deterrence changes what is observed for those projects without changing whether R is saved. The remaining fraction π_D would have succeeded; deterring them newly prevents focal loss. Each newly deterred contest therefore raises expected focal avoided loss by $\pi_D\Lambda_R$. This additional ecological gain tends to zero as defense approaches certainty, although avoiding contests can still save legal, administrative, and political resources.

Contest records, a focal causal evaluation, and a landscape evaluation capture different effects. A contest record, the documented history of challenges to a protected area's status, such as legal

or administrative attempts to downgrade or remove it, counts only contests that occur (box IX of Figure 1). A focal evaluation must also count loss avoided because contests were deterred (box VIII). A landscape evaluation must additionally follow the developer's fallback after R is blocked (box X), since a deterred developer, like a defeated one, can relocate. Deterrence and displacement therefore describe different margins: deterrence determines whether the developer contests protection at R, while displacement describes what happens to the project after R is blocked, whether by deterrence or by a successful defense.

5.2 Displacement and landscape outcomes

The focal accounting asks whether R is saved. The landscape accounting also follows the developer's fallback, which may be to abandon, delay, downsize, redesign, or relocate the project, after either deterrence or defeat. When the fallback is relocation, part of the loss avoided at R reappears elsewhere, the displacement that the impact-evaluation literature calls leakage (Ewers & Rodrigues 2008, Pfaff & Robalino 2017).

Let L^C be expected subsequent landscape damage after protection defeats a contest, and L^D expected damage after a contest is deterred. Each incorporates the probability, scale, timing, and destination of the fallback. Relative to development at the unprotected focal parcel, the landscape effect is

$$A_R^{\text{land}}(S) = \underbrace{c(S)(1-\pi_D)(\Lambda_R - L^C)}_{\text{successful contested defense}} + \underbrace{[\xi - c(S)](\Lambda_R - L^D)}_{\text{deterrence}}. \tag{8}$$

The relevant comparison is not development versus no development, but expected damage at R versus expected damage after R is blocked. The ordering of fallback damage need not be an arbitrary favorable assumption: in a comparable-project setting, it can follow from the rule used to select the protected parcel. Following the operations-research literature, we call a rule that ranks sites purely by ecological value *value-first* targeting: the agency gives priority to the most ecologically valuable land among the sites a developer could use.

Proposition 6 (Value-first targeting and landscape outcomes)**.** *Maintain the accounting in equation* (8)*, with* $0 \le c(S) \le \xi$ *and* $0 < \pi_D < 1$*. Suppose a blocked project is abandoned or*

relocated to one feasible alternative parcel. Suppose also that ecological value ranks the total loss caused by comparable projects, and that the protected parcel R has the highest value in a set containing every feasible relocation destination. Then

$$L^C \leq \Lambda_R, \qquad L^D \leq \Lambda_R, \qquad A_R^{\text{land}}(S) \geq 0.$$

The landscape gain is strictly positive whenever a blocking outcome has positive probability and its expected fallback damage is strictly below Λ_R. The fallback need not be the same after deterrence and defeat.

Both contributions in equation (8) are then non-negative: successful contested defense and deterrence each replace the loss at R with a fallback that is no more damaging. The sign of the landscape effect does not require knowing which of the feasible alternatives the developer will choose. The restriction concerns comparable total development losses: ranking per-hectare ecological quality alone would not suffice if relocation could substantially increase the development footprint. Nor does passing the current-surplus test establish this ordering, because that test also incorporates acquisition costs and readiness benefits.

A rule based on current threat offers no corresponding guarantee. Protecting a threatened but lower-value parcel can redirect development toward a more ecologically valuable site, producing a focal gain but a landscape loss. Threat can therefore inform the urgency of protection without determining which ecological values should be secured. The value-first rule connects the sign of displacement effects to the conservation targeting rule, rather than treating displacement as uniformly harmful or benign.

Whether *additional deterrence* improves the landscape is a separate comparison. If a project has the same fallback after deterrence or defeat, $L^C = L^D = L$, newly deterring a contest that would have been defeated does not change the ecological outcome. Newly deterring one that would have succeeded replaces damage Λ_R with L. The expected landscape gain per newly deterred contest is therefore $\pi_D(\Lambda_R - L)$, which is positive when the fallback is less damaging (Figure 3).

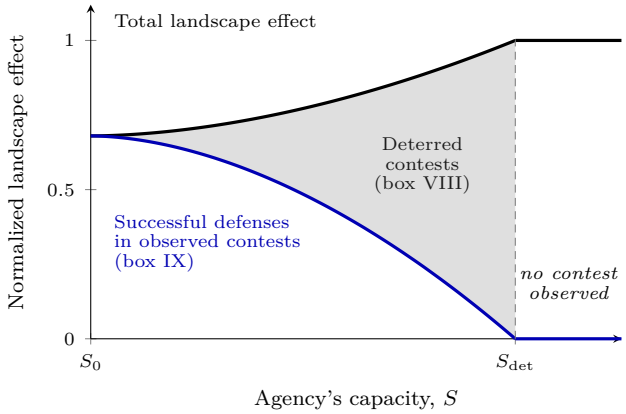


Figure 3: **Deterrence is invisible in contest records.** As the agency's capacity deters contests at R, successful defenses in observed contests (box IX of Figure 1) fall while the total landscape effect rises; the shaded gap is the effect of deterred contests (box VIII). The paths assume that the developer's fallback is the same after deterrence or defeat and less damaging than development at R, $L^C = L^D < \Lambda_R$. The figure is schematic; Online Appendix S.8 records the normalization.

With different fallbacks, stronger deterrence can reduce the landscape gain if deterrence leads to sufficiently more damaging development than defeat. Holding π_D, L^C, and L^D fixed, additional deterrence reduces landscape value precisely when

$$L^D - L^C > \pi_D(\Lambda_R - L^C). \tag{9}$$

Under $L^C \leq \Lambda_R$, a reversal requires deterrence to produce a sufficiently more damaging fallback than successful defense: for example, contest costs or delay might otherwise cause abandonment or downsizing, whereas deterrence sends the intact project elsewhere immediately. Relocation alone does not establish this reversal. Under value-first targeting, stronger deterrence can then reduce the size of the conservation gain without turning protection into a net landscape loss. Online Appendix S.7 gives the derivative and the accounting of outcomes.

Contest formation at R is therefore an insufficient statistic. Evaluation should distinguish parcels never at risk from contests deterred before they are made, and track the scale, timing, and destination of projects after R is blocked. These observations separate avoided loss from relocated harm and prevent the effect of deterrence from being scored as zero.

6 Implications for additionality-based selection and evaluation

Sections 3 and 4 addressed whether a candidate parcel is worth protecting and when to protect it once readiness is counted, showing that protection before a threat arrives can be optimal, and Section 5 showed that protection can also deter contests at R and change where development goes. A third question, central to evaluation, is what protection changed (Table 1). When pre-emption is optimal, these insights complicate the answer to this third question: an evaluation of avoided loss over a stated horizon (outcome additionality) may end before the threat arrives, miss the benefits of readiness at other sites, and, when it relies on records of observed contests (documented attempts to overturn protection), fail to count the losses that deterrence avoids (Section 5). In short, additionality can measure accurately what changed at the protected site over the evaluation horizon, but only under certain conditions does this coincide with the agency's siting and timing priorities. In this section we set out those conditions, and hence when additionality can serve as an eligibility rule for conservation funding.

6.1 Cross-site consistency: when additionality preserves siting priority

The first condition concerns the evaluation horizon. Consider a comparison based only on avoided loss, with effective protection and no cost dynamics or readiness spillovers. A sufficient condition for current-impact rankings to preserve full-horizon rankings is that every parcel's cumulative loss probability be the same positive multiple of its current loss probability: threat must grow in the same proportion at every parcel. A moving frontier need not satisfy this condition.

For example, take two parcels and two periods. The near parcel has ecological value v_N and loss probability $\pi_N > 0$ in the first period, with no further loss risk in the second. The remote parcel has value v_R, no first-period risk, and loss probability $\pi_R > 0$ in the second period. A first-period evaluation credits the near parcel with avoided loss $v_N\pi_N$ and the remote parcel with zero. Over both periods, however, the remote parcel contributes more whenever $v_R\pi_R > v_N\pi_N$. The short evaluation window reverses the ranking simply by ending before the remote parcel is at risk. Cost escalation and readiness can create further differences between measured current impact and the agency's value of protection.

6.2 Within-site consistency: when additionality preserves timing priority

The second concerns timing: an additionality threshold can delay protection. As a development frontier approaches, near-horizon additionality may rise because counterfactual conversion becomes more likely. Acquisition costs, economic and political, can rise at the same time, while the opportunity to build readiness shrinks. A rule that funds protection only once additionality exceeds a threshold can then push protection past the agency's preferred date.

Proposition 7 (Eligibility and the value of protection)**.** *Let $W(x)$ be the value of the opportunity to protect R when the agency follows the optimal policy of Proposition 1. Suppose an eligibility rule restricts when the agency may protect R, for example by requiring an observed additionality measure to reach a threshold first, while leaving information, payoffs, and the risks of losing the opportunity unchanged and still allowing the agency never to protect. Then the agency's optimal value under the rule, $W_E(x)$, satisfies*

$$0 \leq W_E(x) \leq W(x).$$

In the readiness-dependent case $B_R(q_R) < P_R$ with $\kappa_R(q_R, \rho_R) > 0$, let a denote the acquisition threshold. If $x > a$ and every finite eligible protection date is at least some fixed $\varepsilon > 0$, the loss is strict: $W_E(x) < W(x)$.

An eligibility rule removes some of the protection policies available to the agency, so it cannot raise the value of the opportunity. Under the stated conditions, it lowers that value strictly when it imposes a fixed positive minimum delay on protection that is already optimal. The comparison isolates the cost of restricting when the agency can protect, holding funding and everything else fixed; any benefits an eligibility rule may bring, through financing, screening, or incentives, lie outside it. Online Appendix S.3.1 gives the proof; excluding immediate protection alone is not enough for a strict loss if eligible dates can come arbitrarily soon. Low current additionality can reflect low ecological value, ineffective protection, or a horizon that ends before threat arrives. These explanations have different implications for acquisition. Treating them as equivalent can force protection toward the moment when private development pressure is greatest rather than the moment when expected conservation value is highest. Figure 4 illustrates the value lost when eligibility must wait until additionality, $A_R(\tau, H)$ for an evaluator's horizon H, reaches the threshold.

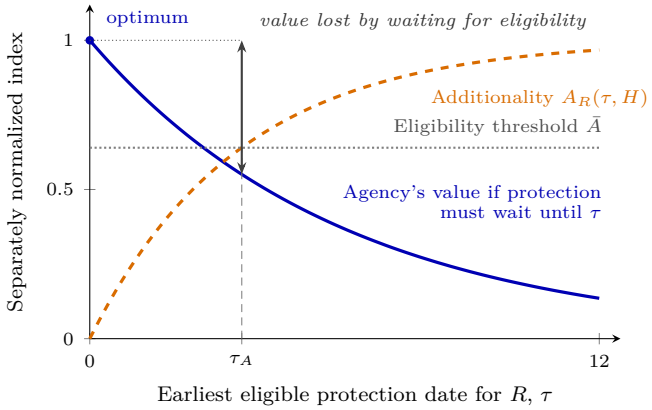


Figure 4: **Eligibility can delay protection.** The figure illustrates the readiness-dependent case, with positive readiness and current stakes above the threshold for immediate protection. A rising additionality score delays eligibility until τ_A; the agency's curve schematically represents the value of the opportunity to protect R when protection cannot occur before τ and the agency can then follow the optimal adaptive policy. The arrow marks the value lost to this eligibility restriction (Proposition 7). The two curves are separately normalized, so only the fall along the agency's own curve is meaningful; their levels are not comparable. The figure is schematic; Online Appendix S.8 records the normalization.

6.3 Evaluation boundaries: readiness across dates and sites

The third condition concerns which sites the evaluation follows. Two protections can have the same effect on conversion at the focal parcel yet different values for the agency if one builds capabilities that are more useful in later conflicts at other sites. A site-level evaluation can correctly estimate its intended causal effect without measuring this institutional spillover; this is a matter of scope, not a flaw in the causal design. An evaluation of Tucabaca's impact confined to the reserve, for example, would not have recorded the committee's part in contesting the clearing at Paquió. Recovering the spillover requires following later interventions, their costs, their success probabilities, or the institutional capacity through which prior protection affects them (box IV of Figure 1). Observing those outcomes matters more than greater precision within a narrow evaluation boundary. The implication is to match the evaluation to the policy objective while retaining a credible counterfactual.

6.4 Instrument boundaries: when additionality is the defining object

In one setting the condition is met by construction: for an avoided-deforestation offset, additionality is part of the definition of the credited environmental service. Groom & Venmans (2023) value offsets through the climate damages they avoid, expressing that value relative to

the social cost of carbon. In their avoided-deforestation model, the correction factor is zero when baseline clearing risk, the risk of clearing without the project, is zero and increases with that risk, holding the other valuation parameters fixed.[4] This is the appropriate valuation for carbon-offset equivalence. The broader decision to acquire land for conservation can nevertheless include benefits and costs outside that component.

To make the connection explicit, write the value of acquisition under current conditions as

$$J_R = V_{\mathrm{AD}}(\tilde{\phi}_R, q_R) + B_R^{\mathrm{other}}(q_R) - P_R(\tilde{\phi}_R) + K\big(x, S, \kappa_R(q_R, \rho_R)\big).$$

Here V_{AD} is the value of the avoided-deforestation benefit underlying carbon crediting, expressed in the same value units as the other terms. The term B_R^{other} includes only direct benefits not already counted in that component, P_R is the effective acquisition cost, and K is the readiness value created by protection, holding the observed stakes, initial capacity, and readiness increment fixed. Carbon equivalence concerns the first term; the agency deciding whether and when to secure the parcel may consider the full sum. This distinction is consistent with Groom and Venmans's recognition of costs and co-benefits beyond the climate benefit itself.

To isolate the acquisition-price channel, hold ecological quality, the remaining offset-valuation parameters, and all readiness inputs fixed, while allowing acquisition cost to rise with baseline clearing risk, $P_R'(\tilde{\phi}_R) > 0$. The avoided-deforestation component then rises with threat, while the rest of acquisition value, $B_R^{\mathrm{other}} - P_R + K$, falls through the price channel. If the increase in acquisition cost exceeds the increase in the avoided-deforestation component, total acquisition value falls even as the carbon-crediting value rises. The two objects can therefore move in opposite directions without either being incorrectly measured.

[4] In their simplest case, with constant hazard rates and constant growth of the social cost of carbon, an immediate project start, and a project horizon that tends to infinity, the correction factor is

$$A_{\mathrm{AD}}(\tilde{\phi}_R) = \frac{r - x_2}{r + \phi - x_2} - \frac{r - x_2}{r + \phi + \tilde{\phi}_R - x_2}.$$

Here $\tilde{\phi}_R$ is the baseline clearing hazard rate, ϕ is the hazard rate of project failure or the ending of additional storage, x_2 is the growth rate of the social cost of carbon, and $r > x_2$ is the discount rate. Thus $A_{\mathrm{AD}}(0) = 0$ and $A_{\mathrm{AD}}'(\tilde{\phi}_R) > 0$.

7 Discussion and conclusion

Conservation priority has a spatial margin and a temporal margin. Siting priority asks which parcels contribute most at a common date; timing priority asks whether a given parcel should be protected now or later. Current threat is relevant to both questions, but once acquisition prices, irreversibility, information, and endogenous institutional readiness are admitted, an ecologically valuable parcel can be important before it is visibly threatened, and worth securing before a short-horizon impact measure reaches its maximum.

The paper identifies two channels that can bring the optimal timing of protection forward. The acquisition-price channel operates through land markets: development expectations can be capitalized into prices before conversion becomes imminent. The readiness channel operates through the conservation agency and its allies: early protection builds legal, administrative, reputational, and political capacity that improves later action. Section 5 adds a separate strategic implication: stronger protected status can change whether a contest forms and what happens to the project after the focal parcel is blocked. This extension also clarifies which effects an evaluation must count; the channels are conceptually distinct and call for different policy instruments.

Remoteness, current threat, and price are state variables, not substitutes for the conservation objective. A remote parcel selected only because it is inexpensive can fail the current-surplus test, exactly as the location-bias literature warns. A high-threat parcel can also be a poor choice if ecological contribution is low, protection is ineffective, or the opportunity cost has already become prohibitive. The relevant question is how each state variable changes the expected value of securing a parcel and the relative value of acting now.

For timing policy, the three groups of Figure 2(a) are informative. Some parcels fail the current-surplus test. Some should be retained as valuable options as development pressure and financing conditions change. Some should be secured immediately because the acquisition window is fragile or because readiness benefits are large. Financial instruments can help implement this distinction. Rights of first refusal, options, revolving funds, bridge finance, and multiyear budgets can preserve a high-value opportunity without treating every candidate as an immediate full acquisition (Yoon & Armsworth 2022, Armsworth et al. 2025), and they are also what allow an agency to time protection to observed conditions (Sections 3.3 and 4.1). The model treats

the agency as able to respond to conditions as they evolve. Budget cycles and funding rules can restrict that flexibility, for instance when acquisitions can be approved only at annual meetings or must be committed in advance; Online Appendix S.6 examines the extreme case in which the protection date is fixed in advance.

We show that the effect of uncertainty on the value of a small addition to readiness depends on initial capacity: under the model's conditions, it is positive for agencies with low capacity and negative for agencies with high capacity (Figure 2(b)). The intuition is clearest near zero capacity, where even a small addition mainly expands access to future intervention. This value comparison does not by itself establish when protection should occur, because uncertainty also changes the value of waiting. But for agencies with sufficiently low initial capacity facing a sufficiently fragile acquisition window, greater uncertainty brings protection forward, for readiness gains of any size. In those circumstances, the result runs against the usual presumption that uncertainty is a reason to wait.

For additionality-based funding, our analysis suggests a division of labor. Additionality should remain central when the instrument promises a quantified unit of avoided loss, as in avoided-deforestation credits. It is also essential for evaluating whether protection changed a specified outcome over a specified horizon. A hard current-additionality threshold is less well suited to selecting a long-run biodiversity portfolio or timing the acquisition of already-identified high-value land. In those settings, the threshold can import the evaluator's horizon into the agency's problem and encourage action only after the acquisition window has closed, while missing readiness benefits at other sites and, where it relies on contest records, the losses that deterrence avoids. The same forces that raise clearing risk also raise the land's price, so, as the comparison with Groom & Venmans (2023) illustrates, rising clearing risk can increase the credited value of a parcel even while lowering the total value of acquiring it. A practical design would report ecological or portfolio contribution, dynamic timing value, and expected outcome additionality separately before combining them under an explicit objective.

The analysis is deliberately single-parcel: it counts the readiness spillover directly through K but takes the surrounding landscape as given. Section 5 shows that, in a comparable-project setting, value-first targeting (giving priority to the most ecologically valuable land among the sites a developer could use) ensures that a blocked developer's fallback is no more damaging than development at the protected parcel. Under these conditions, protection of a parcel cannot

increase expected damage across the landscape, because blocked development is either abandoned or relocated to land where it does no more ecological damage. A corollary is that conservation targeting governs both which parcels are worth securing and whether the development they displace does less harm elsewhere. The model complements Weinhold & Andersen (2026), who analyze siting priority within a sequential adversarial game without such spillovers; endogenizing where development relocates jointly with what is protected would require a portfolio problem and is a natural extension.

In sum, the paper reframes conservation priority as two decisions rather than one: which parcels are worth securing, and when. For practitioners and conservation finance, this yields concrete guidance: value a candidate for the biodiversity it protects, for what it costs, and for the institutional capacity that securing it builds for future conservation elsewhere. More broadly, well-targeted protection can do two things at once: secure the parcels that matter and, when blocked development shifts toward less damaging land, improve the wider landscape rather than merely relocate harm. Empirically estimated additionality should inform these choices, but on its own a short-horizon, site-level measure is an incomplete rule for deciding what to protect or when. Our illustrative example of early protection at Tucabaca shows the mechanism: conservation that residents sought for their water also built a committee that later helped contest clearing in a neighboring reserve. Ultimately, protection is a defense of land and an investment in the capacity to defend it: conservation today can preserve land and also improve the institution that will defend nature tomorrow. The value of local action can thus lie partly in the options it makes possible for the future.

Appendix

Appendix A collects the core notation and Appendix B gives overviews of the proofs; the derivations and full proofs are in the Online Supplementary Appendix, available at https://dmweinhold.github.io/Latest-Version-Papers/Weinhold_Readiness_online_supplement.pdf.

A Notation

Table 2: Core model notation

Symbol	Definition	Symbol	Definition
Portfolio and focal parcel			
R	Focal candidate parcel	C	Later conflict at another site
ν	Opportunity cost of a common conservation resource	d_R; c_R	Direct burden; use of the common resource
$B_R(q_R)$	Shorthand for marginal direct portfolio contribution	P_R	Shorthand for current effective acquisition cost
$D_R(q_R)$	Direct net value, $B_R(q_R) - P_R$	$L_R(q_R)$	Direct shortfall, $P_R - B_R(q_R)$
SV_R	Current surplus, $B_R(q_R) - P_R + K$	$K(x, S, \kappa)$	Readiness value created by protecting R, equation (1)
Readiness, timing, and the later conflict			
S_t; S	Institutional capacity; current capacity	$\kappa_R(q_R, \rho_R)$	Readiness increment created by protecting R
ρ_R	Relevance of experience at R to later conflicts	δ	Readiness depreciation rate
T_R; γ_R	Time the acquisition window closes; its hazard rate	T_C; λ	Arrival time of the later conflict; its hazard rate
h_R	Combined hazard rate, $\gamma_R + \lambda$	r	Discount rate / opportunity cost of commitments
X_t; x	Stakes at C; their current level	μ; σ	Drift and volatility of X_t
$I(S)$	Cost of intervening at C	$p(S)$	Probability that intervention at C is feasible (access)
$F(x; I)$	Value of the option to intervene at C; its continuation value below $x^*(S)$	$\mathcal{F}(x, S)$	Access-adjusted option value
$\Omega(x, S)$	Marginal value of readiness, $\partial K/\partial\kappa$ at $\kappa = 0$	β	Positive characteristic root for F
τ	Protection date	q^0; q^*	Current-surplus and immediate-protection thresholds
Q	Readiness contribution to immediate protection, $K - xK_x/\alpha$	α	Positive root of $\frac{1}{2}\sigma^2\alpha(\alpha-1) + \mu\alpha = r + \lambda + \gamma_R$
$x^*(S)$	Intervention threshold at C	S^*	Capacity at which uncertainty's effect on the marginal value of readiness changes sign
a	Stakes at which protection of R becomes optimal	$\bar{\gamma}_R$; $\bar{S}$	Window-closure rate and capacity bounds in Proposition 4 and Corollary 1
ψ	Sensitivity of access to readiness in the exponential case	I_0; χ	Intervention cost at zero capacity; proportional cost-reduction rate in the exponential case
$\chi_I(S)$	General proportional cost-reduction rate, $-I'(S)/I(S)$		
W_t	Standard Brownian motion	$W(x)$	Optimal value of the opportunity to protect R

Notes: The table records the core model notation. Evaluation-specific quantities in Section 6 – including additionality A_R, the evaluation horizon H, and the deterrence and fallback variables n, θ, π_D, ξ, Λ_R, L^C, and L^D – are defined where introduced, in Section 5 and Online Appendix S.7. Arguments held fixed are suppressed for readability. For a candidate of quality q_R, readiness value is $K(x, S, \kappa_R(q_R, \rho_R))$; the current state and relevance parameters, together with the existing portfolio and resource opportunity cost, are held fixed as described in Section 3.2.

B Proof overviews

This appendix gives the intuition behind the main results; the full proofs are in the Online Supplementary Appendix.[5]

Proposition 1. If direct benefit covers acquisition cost, delay can only lose value, so immediate protection is optimal. Otherwise, and when protection creates positive readiness, choosing when to protect is an optimal stopping problem: readiness value rises with the stakes at C, and the opportunity to protect is lost at the combined rate of window closure and conflict arrival. The readiness contribution Q can be written as a weighted average of terms that rise with the stakes, so the condition $B_R + Q \geq P_R$ holds above a single threshold. A standard verification argument then shows that protecting as soon as the stakes reach that threshold is optimal among all protection policies, and that $0 < Q < K$ (Online Appendix S.2).

Proposition 2. Higher future prices lower the payoff of every policy that protects later and leave the payoff from immediate protection unchanged. Comparing policies one at a time gives the result, without solving the problem with changing prices (Online Appendix S.3.2).

Proposition 3. The marginal value of readiness splits into an access part and a cost-reduction part. Its response to uncertainty has the sign of the difference between two terms. The first measures the relative strength of access gains; it is very large at low capacity and falls toward zero as capacity rises. The second measures how sensitive the option's response to uncertainty is to the intervention cost; because that response is log-concave as a function of $\log(x/I)$, this term rises with capacity. The difference therefore changes sign exactly once (Online Appendix S.4).

Proposition 4 and Corollary 1. As the acquisition window becomes more fragile, the adjustment that separates Q from K vanishes, so Q and its response to uncertainty converge to readiness value and its response. A strictly positive response of readiness value at the current-surplus threshold therefore carries over to the threshold for immediate protection once the window is fragile enough. At zero capacity, protection supplies access to an option whose

[5] Available at https://dmweinhold.github.io/Latest-Version-Papers/Weinhold_Readiness_online_supplement.pdf.

value rises with uncertainty and there is no existing option value to subtract, so readiness value responds positively; continuity extends this to low capacity (Online Appendix S.5).

Propositions 5 and 6. Proposition 5 follows directly from the accounting in equation (7), whose outcomes are set out in Online Appendix S.7. For Proposition 6, value-first targeting makes the loss at every feasible relocation destination no larger than the loss at R, so expected fallback damage after either deterrence or defeat, an average of zero damage from abandonment and those losses, is no larger either. Both terms of equation (8) are therefore non-negative (Online Appendix S.7).

Proposition 7. The result follows because an eligibility rule only removes protection policies, leaving payoffs and information unchanged. When immediate protection is strictly optimal in the readiness-dependent case, the value of the opportunity falls strictly while protection is forced to wait, so a fixed minimum delay produces a strict loss (Online Appendix S.3.1).

C AI-use disclosure

Generative AI tools were used during manuscript preparation. OpenAI's ChatGPT Pro contributed to mathematical modeling, formal proof development, and structural editing; Anthropic's Claude Opus contributed to proof checking and structural editing. AI-assisted derivations and numerical results were cross-checked independently, and the authors reviewed all content. The authors take sole responsibility for all claims, all errors, and all editorial decisions made in the paper.

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